L_0 regularization for subnational microsimulation calibration

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Abstract

Tax-benefit microsimulation models typically operate at the national level, using household survey weights calibrated to aggregate population targets. Subnational analysis—at the level of states, congressional districts, or local authorities—requires datasets that simultaneously satisfy geographic distributional constraints while preserving household-level detail. We present a method based on L_0 regularization that jointly optimizes survey weight magnitudes and sparsity to produce calibrated subnational microsimulation datasets.

Our approach builds on the Hard Concrete distribution (Louizos et al., 2018), which induces exact sparsity by multiplying each household's weight by a learned stochastic gate that collapses to a deterministic zero or one at inference time. We parameterize each gate with a log-alpha and temperature parameter, and jointly optimize these alongside log-transformed weight magnitudes using a single loss function combining scale-invariant relative calibration error, an L_0 sparsity penalty on the expected count of active households, and a light L_2 regularizer on weight magnitudes.

The pipeline begins with the US Current Population Survey. Each household record is cloned multiple times and assigned to random census blocks drawn from a population-weighted distribution. Program participation indicators are re-randomized per geographic assignment using local take-up rates. Each clone is then run through POLICYENGINE's tax-benefit microsimulation engine to generate geography-specific outputs. The L_0 optimizer selects which household-geography combinations to retain, calibrating simultaneously against approximately 37,800 targets across three geographic levels. The sparsity penalty is configurable: a higher penalty produces a compact national dataset of approximately 50,000 records, while a lower penalty yields a larger dataset of approximately 3–4 million records covering all 436 congressional districts and 50 states individually. The method is implemented as the open-source `l0-python` PyTorch package.

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1 Introduction

Microsimulation models estimate the effects of tax and benefit policies on households by applying program rules to individual-level microdata. Most operational models—including those maintained by the Congressional Budget Office (Congressional Budget Office, 2018), the Joint Committee on Taxation (Joint Committee on Taxation, 2023), and the Tax Policy Center (Tax Policy Center, 2024)—operate at the national level. They calibrate household survey weights to aggregate administrative totals such as total income tax revenue, program enrollment counts, and demographic benchmarks, then use the reweighted dataset to simulate policy reforms.

Subnational policy analysis introduces a fundamentally different calibration challenge. Rather than matching a single set of national aggregates, the microdata must simultaneously reproduce distributional statistics at multiple geographic levels: congressional districts, states, and the nation as a whole. A dataset calibrated for the state of California must match California-specific IRS income totals, SNAP participation counts, Medicaid enrollment, and age distributions, while remaining consistent with national budget projections from the CBO and tax expenditure estimates from the JCT. Across 436 congressional districts and 50 states, this produces approximately 37,800 simultaneous calibration targets.

Existing calibration methods scale poorly to this setting. Iterative proportional fitting (IPF; Deming and Stephan, 1940; Ireland and Kullback, 1968) adjusts weights along one dimension at a time, cycling through marginal constraints until convergence. IPF handles cross-classified tables but does not naturally accommodate hierarchical geographic constraints—district targets must sum to state targets, which must sum to national targets—without ad hoc post-processing. Generalized regression (GREG) estimators (Deville and Särndal, 1992; Särndal, 2007) solve a constrained optimization problem that minimizes distance from initial weights subject to exact calibration constraints. GREG produces a closed-form solution for moderate numbers of constraints but becomes computationally intractable and numerically unstable as the constraint count approaches the tens of thousands.

Spatial microsimulation methods take a different approach, often distinguishing between reweighting methods and synthetic reconstruction methods for constructing small-area microdata (Tanton, 2014). Within this broader literature, researchers have used combinatorial optimization and simulated annealing (Williamson et al., 1998; Huang and Williamson, 2001; Harland et al., 2012) as well as deterministic reweighting (Tanton et al., 2011; Lovelace

and Dumont, 2016). These methods typically operate at a single geographic level and require separate calibration runs for each area, making joint multi-level calibration difficult.

This paper presents a method that addresses these limitations by jointly optimizing weight magnitudes and sparsity in a single gradient-based framework. We adapt the Hard Concrete distribution (Louizos et al., 2018), originally developed for neural network pruning, to the survey calibration setting. Each household-geography combination receives a continuous weight and a stochastic binary gate. The gate is parameterized by a learnable logit and trained via gradient descent to minimize a loss function that combines relative calibration error across all 37,800 targets with an L_0 penalty on the expected number of active records. At inference time, the stochastic gates collapse to deterministic zeros and ones, producing a sparse dataset in which most household-geography combinations are dropped while the retained records carry calibrated positive weights.

The approach builds on Woodruff and Ghenis (2024), who developed a two-stage methodology for constructing enhanced national microsimulation datasets from the Current Population Survey (CPS) and the IRS Public Use File (PUF). Their method uses quantile regression forests (QRF) to impute 72 tax variables from the PUF onto CPS records, then applies dropout-regularized gradient descent to reweight the combined dataset against approximately 7,000 national targets. The present paper extends this framework from a single national dataset to subnational coverage by introducing three new components: (a) a clone-and-assign procedure that replicates each CPS household across multiple geographic locations, (b) L_0 Hard Concrete gates that replace dropout regularization and enable exact sparsity, and (c) a hierarchical uprating scheme that reconciles targets from different administrative sources at district, state, and national levels. Because the method still solves a calibration-weighting problem over survey-based microdata, GREG and IPF are the closest classical empirical comparators; at the same time, the clone-and-assign pipeline produces derived spatial microdata files that are also relevant to synthetic-population use cases.

The configurable sparsity penalty produces datasets of different sizes for different use cases. A high penalty ($\lambda_{L_0} = 10^{-4}$) retains approximately 50,000 records, suitable for national-level web-based simulation where download size and computation time matter. A low penalty ($\lambda_{L_0} = 10^{-8}$) retains approximately 3–4 million records, preserving geographic resolution for all 436 congressional districts.

The method is implemented in open-source Python packages and deployed as part of POLICYENGINE's US microsimulation model (`https://policyengine.org/us/model`), which uses these calibrated datasets for both national and subnational policy analysis.

The remainder of this paper is organized as follows. Section 2 reviews survey cali-

bration methods, spatial microsimulation, and the Hard Concrete distribution. Section 3 describes the input microdata and calibration target sources. Section 4 details the four-stage pipeline: clone creation, calibration matrix construction, L_0 optimization, and dataset assembly. Section 5 presents calibration accuracy, sparsity analysis, and comparisons with IPF and GREG. Section 6 discusses trade-offs, limitations, and generalizability. Section 7 concludes.

2 Background and related work

2.1 The survey calibration problem

Given a probability sample of n units with initial design weights d_i and a vector of auxiliary variables $\mathbf{x}_i \in \mathbb{R}^p$ for each unit i, survey calibration seeks adjusted weights w_i such that the weighted sample totals reproduce known population totals:

$$\sum_{i=1}^{n} w_i \mathbf{x}_i = \mathbf{T} \tag{1}$$

where $\mathbf{T} \in \mathbb{R}^m$ is the vector of m known administrative or census totals. When $m < n$, the system is underdetermined and a distance criterion selects among feasible solutions. Deville and Särndal (1992) formalized this as a constrained optimization problem: minimize a distance function $\sum_i G(w_i, d_i)$ subject to Equation 1, where G is convex and satisfies $G(d_i, d_i) = 0$. The choice of G determines the calibration method: a chi-squared distance yields the generalized regression (GREG) estimator, an exponential distance yields raking (multiplicative calibration), and a truncated distance yields bounded weight adjustments. Särndal (2007) provides a comprehensive review of these methods and their statistical properties.

In the subnational setting addressed here, the constraint vector $\mathbf{T}$ contains targets at multiple geographic levels—congressional districts, states, and the nation—requiring that district totals sum to state totals, which sum to national totals. This hierarchical structure produces $m \approx 37{,}800$ simultaneous constraints with near-collinearity across levels, placing the problem beyond the regime where classical closed-form calibration methods operate reliably.

Subnational microsimulation also sits within the broader spatial microsimulation literature. Reviews of static spatial microsimulation often distinguish between *reweighting* methods, which begin from survey microdata and adjust weights to match small-area con-

straints, and *synthetic reconstruction* methods, which construct new small-area populations from aggregate tables (Tanton, 2014; O'Donoghue et al., 2014). This distinction matters for the present paper. Our method remains, at core, a calibration-weighting approach over observed CPS households, so GREG and IPF are the closest classical benchmarks. At the same time, by cloning households, assigning them to new geographies, and assembling area-specific output files, the pipeline also produces derived spatial microdata for small-area policy analysis, making it relevant to some synthetic-population use cases even though the empirical comparison in this paper focuses on calibration methods.

2.2 Generalized regression (GREG) estimation

The GREG estimator minimizes the chi-squared distance $\sum_i (w_i - d_i)^2/d_i$ subject to Equation 1, yielding the closed-form solution:

$$w_i = d_i + d_i\,\mathbf{x}_i^\top \left(\sum_{j=1}^{n} d_j\,\mathbf{x}_j\mathbf{x}_j^\top\right)^{-1} \left(\mathbf{T} - \sum_{j=1}^{n} d_j\,\mathbf{x}_j\right) \tag{2}$$

This requires inverting a $p \times p$ matrix, where p is the dimension of the auxiliary vector. When p is moderate (tens to hundreds), GREG is computationally efficient, statistically well understood, and produces asymptotically unbiased estimators (Deville and Särndal, 1992; Särndal, 2007).

GREG has two practical limitations in the subnational setting. First, the matrix inversion in Equation 2 becomes numerically ill-conditioned as p grows into the thousands, particularly when constraints at different geographic levels are nearly collinear (e.g., the sum of district age counts within a state nearly equals the state age total). Second, GREG does not constrain weight signs: calibrated weights can become negative, which is undesirable for microsimulation since negative weights imply subtracting households from the population. Weight truncation or bounding after the fact restores non-negativity but violates the calibration constraints.

The Congressional Budget Office (Congressional Budget Office, 2018), Joint Committee on Taxation (Joint Committee on Taxation, 2023), and Tax Policy Center (Tax Policy Center, 2024) use variants of GREG calibration for their national microsimulation models, where the number of calibration constraints is typically in the hundreds—well within GREG's reliable operating range.

2.3 Iterative proportional fitting

Iterative proportional fitting (IPF; Deming and Stephan, 1940; Ireland and Kullback, 1968) adjusts cell counts in a contingency table to match given marginal totals. The algorithm cycles through dimensions, scaling each dimension's cells so that their marginal matches the target, then repeating until convergence. IPF converges to the maximum entropy solution subject to the marginal constraints (Ireland and Kullback, 1968). In the spatial microsimulation literature, it appears in both synthetic reconstruction and reweighting forms; the discussion here concerns the reweighting form that starts from survey microdata and updates weights rather than building a synthetic joint distribution from scratch (Tanton, 2014).

In the microsimulation context, IPF adjusts household weights to match cross-classified population counts—for example, persons by age group within each congressional district. IPF has several practical advantages: it preserves non-negativity by construction (weights are scaled multiplicatively, so positive weights remain positive), it requires no matrix inversion, and it scales well to high-dimensional contingency tables. EUROMOD, the EU-wide tax-benefit microsimulation model, uses IPF-based calibration to reweight national surveys to demographic benchmarks across member states.

However, IPF has three limitations relevant to subnational calibration. First, IPF does not naturally enforce hierarchical consistency: district-level targets produced by IPF do not automatically sum to the correct state totals, requiring post-hoc reconciliation that may introduce new inconsistencies. Second, IPF handles only count targets organized as contingency table margins; incorporating continuous-valued targets (e.g., aggregate income or benefit spending) requires auxiliary procedures outside the IPF framework. Third, IPF scales cell counts multiplicatively, which can produce extreme weights when initial cells are small or zero, and convergence slows or fails when marginal constraints are mutually inconsistent.

2.4 Spatial microsimulation

Spatial microsimulation constructs small-area populations either by reweighting existing microdata or by synthesizing new unit-record populations from aggregate constraints (Tanton, 2014; O'Donoghue et al., 2014). Williamson et al. (1998) introduced a combinatorial optimization approach that selects a subset of survey records for each small area using simulated annealing to minimize the difference between weighted survey totals and census benchmarks. Huang and Williamson (2001) extended this with a deterministic algorithm based on systematic record selection. Tanton et al. (2011) applied generalized regression

reweighting to create small-area estimates of poverty and housing stress in Australia.

Harland et al. (2012) developed methods for creating realistic synthetic populations at fine geographic scales using iterative proportional fitting combined with Monte Carlo sampling. Lovelace and Dumont (2016) provided an accessible implementation in R with the `spatial-microsim-book` framework.

Within this literature, combinatorial optimization and especially simulated annealing occupy an important place as methods for generating synthetic spatial microdata from observed survey records (Tanton, 2014; O'Donoghue et al., 2014). Their main advantage is that they work directly with real microdata, can be flexible about household structure, and can accommodate settings where the unit of analysis in the constraints and the microdata do not align neatly. Their main disadvantage is computational intensity: they are usually run area by area and search a large discrete space of candidate record combinations (Harland et al., 2012; O'Donoghue et al., 2014).

These methods typically operate at a single geographic level—producing estimates for each small area independently. Joint calibration across multiple geographic levels (district, state, national) with a single set of weights is uncommon in the spatial microsimulation literature, as it requires simultaneously satisfying tens of thousands of constraints that span different administrative geographies. Other operational models avoid the problem entirely: TAXSIM (NBER) operates at the national level without geographic calibration, while state-level models maintained by individual state revenue departments calibrate only within their own jurisdiction. For this reason, simulated annealing is an important reference point in the broader spatial microsimulation literature, but it is not the closest like-for-like empirical comparator to our setting. The benchmark design in this paper therefore focuses on GREG and IPF as the classical calibration baselines most closely aligned with a shared weighted-microdata formulation.

2.5 L_0 regularization and the Hard Concrete distribution

L_0 regularization penalizes the count of nonzero parameters in a model, encouraging sparsity. Unlike L_1 regularization, which penalizes the sum of absolute values and produces approximate sparsity through shrinkage, L_0 regularization produces exact zeros. The L_0 norm $\|\mathbf{w}\|_0 = \sum_i \mathbf{1}[w_i \neq 0]$ is discontinuous and combinatorial, making direct optimization intractable for large parameter spaces.

Louizos et al. (2018) proposed a differentiable relaxation of L_0 regularization using the Hard Concrete distribution. Each parameter is multiplied by a stochastic gate z_i drawn from

a stretched and rectified Bernoulli distribution. During training, the gate is sampled as:

$$u \sim \text{Uniform}(0, 1) \tag{3}$$

$$\bar{s} = \sigma\left(\frac{\log u - \log(1-u) + \log \alpha_i}{\beta}\right) \tag{4}$$

$$s = \bar{s}(\zeta - \gamma) + \gamma \tag{5}$$

$$z_i = \min(1, \max(0, s)) \tag{6}$$

where σ is the sigmoid function, α_i is a learnable parameter controlling the gate's probability of being open, β is a temperature parameter, and $\gamma < 0$ and $\zeta > 1$ are stretch parameters that place probability mass at exactly zero and exactly one after clipping. The expected L_0 penalty is:

$$\mathbb{E}[L_0] = \sum_i \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right) \tag{7}$$

which is differentiable with respect to α_i and can be optimized with standard gradient methods.

At inference time, the stochastic sample is replaced by the deterministic mean:

$$z_i^{\text{det}} = \min(1, \max(0,\ \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma)) \tag{8}$$

The Hard Concrete distribution was originally developed for pruning neural network weights (Louizos et al., 2018). In the present work, we repurpose it for survey weight selection: each household-geography combination receives a gate, and the L_0 penalty controls how many combinations are retained in the final dataset.

2.6 National-level predecessor

Woodruff and Ghenis (2024) developed a two-stage method for constructing enhanced national microsimulation datasets. In the first stage, quantile regression forests (Meinshausen, 2006) impute 72 tax variables from the IRS Public Use File onto CPS records, preserving joint distributional characteristics through sequential conditioning. In the second stage, dropout-regularized gradient descent optimizes household weights to match approximately 7,000 national-level administrative targets from the IRS, CBO, JCT, Census Bureau, and program-specific sources.

The present paper extends this framework in four directions. First, we replace the single-geography dataset with a clone-and-assign procedure that replicates each CPS house-

hold across 436 congressional districts, 50 states (plus DC), and New York City, enabling subnational coverage. Second, we replace dropout regularization with L_0 Hard Concrete gates, providing a principled mechanism for exact sparsity with a configurable sparsity-accuracy trade-off. Third, we introduce hierarchical uprating to reconcile targets from different administrative sources—where, for example, district-level IRS totals may not sum exactly to state-level USDA program totals due to different data collection methodologies and reference periods. Fourth, we add take-up re-randomization so that program participation varies across geographic clones of the same household, reflecting local-area take-up rates.

3 Data

The calibration pipeline consumes two categories of input: base microdata from household surveys, and administrative targets from government agencies. This section describes both.

3.1 Base microdata

3.1.1 Current Population Survey

The primary microdata source is the Current Population Survey Annual Social and Economic Supplement (CPS ASEC; U.S. Census Bureau, 2024), a nationally representative household survey of approximately 200,000 persons conducted jointly by the US Census Bureau and the Bureau of Labor Statistics. The CPS ASEC contains detailed demographic variables (age, sex, race, household composition, state of residence), income variables (earnings, Social Security, pension, unemployment compensation), and program participation indicators (SNAP, Medicaid, SSI, TANF).

The CPS has well-documented limitations for tax microsimulation. Top incomes are truncated for privacy protection, causing underestimation of capital gains, partnership income, and other high-income components (Burkhauser et al., 2012). Benefit receipt is underreported relative to administrative totals (Rothbaum and Bee, 2021; Meyer et al., 2021). Tax-specific variables—itemized deductions, tax credits, and filing-status-dependent calculations—are either absent or imprecisely measured.

3.1.2 IRS Public Use File imputation

To address these gaps, we impute 72 tax variables from the IRS Public Use File (PUF; Bryant, 2023) onto CPS records following the methodology of Woodruff and Ghenis (2024).

The PUF contains individual tax return data with detailed income components, deductions, and credits, but lacks household structure and demographic detail.

The imputation uses quantile regression forests (QRF; Meinshausen, 2006) trained on the PUF and applied to CPS records. To preserve the joint distribution across the 72 variables, QRF models are applied sequentially: each variable is imputed conditional on all previously imputed variables in the sequence. A stratified subsample of PUF training records preserves the top 0.5% of the AGI distribution (retaining all records above the 99.5th percentile) alongside a random sample of 20,000 records from the remainder.

The imputation overwrites a subset of 44 unreliable CPS variables with PUF predictions, while retaining original CPS values for all other variables. Each CPS record receives QRF-imputed values for the 72 tax variables, producing an enhanced dataset that combines the demographic detail of the CPS with the tax fidelity of the PUF.

Social Security income, which appears as a single total in the PUF, is decomposed into four sub-components (retirement, disability, survivors, dependents) using a QRF model trained on CPS records where all four components are observed. An age-based heuristic serves as a fallback: records with age 62 or above receive 100% retirement allocation; younger records receive 100% disability allocation.

3.1.3 Source imputation from additional surveys

Nine additional variables are imputed from three external surveys to fill gaps in CPS measurement:

- **American Community Survey (ACS) 2022**: rent and real estate taxes, imputed with state FIPS as a predictor. Because geography is assigned before source imputation (Section 4.1), the ACS QRF conditions on the clone's assigned state, producing state-aware housing cost predictions.
- **Survey of Income and Program Participation (SIPP) 2023**: tip income, bank account assets, stock assets, and bond assets. SIPP public-use files lack state identifiers, so these imputations are state-blind.
- **Survey of Consumer Finances (SCF) 2022**: net worth, auto loan balance, and auto loan interest. SCF also lacks state identifiers; imputations are state-blind.

Training sample sizes are 10,000 household heads for ACS, up to 20,000 weighted-probability records for SIPP, and 50% of SCF records.

3.2 Calibration targets

The optimizer calibrates against approximately 37,800 targets spanning three geographic levels and multiple administrative sources. Table 1 summarizes the target structure.

Source	Targets	Geographies
Congressional district level		
IRS SOI (AGI, income tax, EITC, 21 income/ded. domains)	25,288	436 CDs
Census ACS S0101 (person count by 18 age bands)	7,848	436 CDs
Census ACS S2201 (SNAP household count)	436	436 CDs
State level		
IRS SOI (same domains as district)	2,958	51 states
Census ACS S0101 (person count by 18 age bands)	918	51 states
USDA FNS SNAP (spending + household count)	102	51 states
CMS Medicaid (enrollment)	51	51 states
Census STC (state income tax collections)	51	51 states
National level		
IRS SOI (domain-constrained aggregates, EITC)	51	1
Curated totals (CBO, JCT, SSA, CMS, Census)	37	1
Census ACS S0101 (person count by 18 age bands)	18	1
Total	**37,758**	

Table 1: Calibration targets by geographic level and source. District targets dominate by count. Full variable listing in Appendix B.

Source: `policy_data.db` active target count.

3.2.1 Congressional district targets

At the district level (436 congressional districts), targets come from two sources:

- **IRS Statistics of Income (SOI)**: adjusted gross income, income tax, EITC by number of qualifying children, capital gains, self-employment income, pension income, and other income and deduction categories. These are available at the ZIP code level and aggregated to congressional districts.
- **Census ACS**: age distributions in 18 bands (0–4, 5–9, . . . , 80–84, 85+), providing demographic structure within each district.

- **SNAP household counts**: district-level counts of SNAP-receiving households from USDA administrative data.

3.2.2 State targets

At the state level (50 states plus DC), additional targets include:

- **USDA**: administrative SNAP spending by state.
- **CMS**: Medicaid enrollment by state.
- **Census Bureau**: state income tax collections.

3.2.3 National targets

National targets provide top-level fiscal and demographic constraints:

- **CBO**: federal budget projections for program spending and revenue.
- **JCT**: tax expenditure estimates for major deductions and credits.
- **SSA**: benefit totals by program (retirement, disability, survivors, SSI).
- Additional administrative totals for healthcare spending and demographic benchmarks.

3.3 Hierarchical uprating

Calibration targets originate from different sources with different reference periods. IRS SOI district data may reference tax year 2022, while USDA SNAP data references fiscal year 2024 and CBO projections reference the current budget year. Two adjustments reconcile these differences.

The *uprating factor* (UF) bridges the time gap between the source data period and the calibration year. For most domains, dollar-denominated targets use the Consumer Price Index and count targets use Census population growth projections. For ACA premium tax credits, state-specific uprating factors derived from CMS/KFF enrollment data capture state-level variation in marketplace enrollment growth.

The *hierarchy inconsistency factor* (HIF) corrects for district-level totals that do not sum exactly to state-level totals, which occurs when district and state data come from different administrative sources or use different collection methodologies. For each state s and

variable v, the HIF is:

$$\text{HIF}_{s,v} = \frac{T_{s,v}^{\text{state}}}{\sum_{d \in s} T_{d,v}^{\text{district}}} \tag{9}$$

where $T_{s,v}^{\text{state}}$ is the state-level target and $T_{d,v}^{\text{district}}$ is the district-level target for district d within state s. Each district target is then adjusted as $T_{d,v}^{\text{adj}} = T_{d,v}^{\text{district}} \times \text{HIF}_{s,v} \times \text{UF}_{s,v}$, ensuring that adjusted district targets sum exactly to the uprated state total:

$$\sum_{d \in s} T_{d,v}^{\text{adj}} = \text{UF}_{s,v} \times T_{s,v}^{\text{state}} \tag{10}$$

For IRS SOI data, where district totals are constructed from ZIP-code-level data that sums exactly to state totals, the HIF equals 1.0. For SNAP household counts, where district estimates from one source substantially undercount state administrative totals from another, HIFs range from 1.2 to 1.7 across states.

4 Methodology

The calibration pipeline transforms raw CPS microdata into a calibrated subnational dataset through four stages: (1) clone creation and geographic assignment, (2) calibration matrix construction, (3) L_0-regularized weight optimization, and (4) dataset assembly. Figure 1 provides an overview.

[Pipeline overview diagram to be inserted]

Figure 1: Four-stage calibration pipeline. Stage 1 clones CPS records and assigns geography. Stage 2 builds the sparse calibration matrix by simulating each clone through the tax-benefit engine. Stage 3 runs the L_0 optimizer to find calibrated weights with configurable sparsity. Stage 4 expands the flat weight vector into per-area H5 dataset files.

4.1 Stage 1: clone creation and geographic assignment

The base CPS contains approximately 100,000 household records, each with a single state of residence. A stratified subsampling step reduces this to approximately 12,000 households, preserving the top 0.5% of the AGI distribution in full while uniformly sampling from the remainder. This reduction makes downstream microsimulation computationally feasible while retaining distributional diversity.

To cover 436 congressional districts and 50 states, we replicate each of the approximately 12,000 households N times ($N = 430$ in the default configuration) and assign each clone a random census block from a population-weighted distribution. Although the clone count is slightly below the number of congressional districts, every record is replicated 430 times across different geographic areas, providing the optimizer with sufficient degrees of freedom to place each household type wherever targets require it.

4.1.1 Block sampling

Census blocks are the finest geographic unit in the decennial census. Each block maps deterministically to a congressional district, county, tract, and state. The sampling distribution $P_{\text{pop}}(\text{block})$ is proportional to the block's share of the national population. Drawing blocks rather than congressional districts ensures fine-grained geographic variation within districts and enables derivation of county-level variables (Section 4.4).

4.1.2 AGI-conditional routing

High-income households are routed toward high-AGI congressional districts using a modified sampling distribution. Let A_d denote the aggregate adjusted gross income target for district d from IRS SOI data, and let $P_{\text{pop}}(b)$ denote the population-weighted block probability for block b in district d. For households with AGI above the 90th percentile, the sampling distribution is:

$$P_{\text{AGI}}(b) \propto P_{\text{pop}}(b) \cdot A_{d(b)} \tag{11}$$

where $d(b)$ is the congressional district containing block b. This makes blocks in high-AGI districts more likely targets for high-income households, improving the initial geographic allocation before optimization. Without AGI-conditional routing, the optimizer would need to zero out high-income records that landed in low-AGI districts by chance, wasting capacity for population and demographic targets in those districts.

4.1.3 No-collision constraint

The same CPS household must not appear in the same congressional district in two different clones. If household h in clone c_1 lands in district d, then household h in clone c_2 must land in a different district. This prevents a high-weight household from dominating a small district's targets. The constraint is enforced by resampling: clone 0 draws blocks freely; each subsequent clone checks for collisions against all previous clones and resamples colliding

records for up to 50 retries. Residual collisions after 50 retries are accepted; these are rare with 430 clones and the large block distribution.

After cloning, the total number of household-geography combinations is $N_{\text{clones}} \times n_{\text{records}}$. With 430 clones and approximately 12,000 base households (after stratified subsampling), this produces approximately 5.2 million column positions in the calibration matrix.

4.2 Stage 2: calibration matrix construction

The calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$ maps n household-geography combinations (columns) to m calibration targets (rows). Entry M_{ji} is the contribution of record i to target j—for dollar targets, this is the household's simulated value of the target variable; for count targets, it is 1 if the household satisfies the target's domain constraint and 0 otherwise.

4.2.1 Per-state parallel simulation

The matrix is populated by running each household through POLICYENGINE's tax-benefit microsimulation engine. Because many target variables depend on state-specific tax and benefit rules, a separate simulation is required for each state. A parallel dispatcher sends one job per unique state FIPS code to a pool of worker processes. Each worker creates a fresh `Microsimulation` instance, overwrites every household's `state_fips` with the target state, invalidates cached downstream variables, and calculates all target variables at the household and person levels, accounting for differences in state legislation.

For county-dependent variables—currently limited to ACA premium tax credits, where eligibility depends on county-level benchmark plan premiums—an additional simulation loop iterates over counties within each state, setting the county FIPS for each iteration.

4.2.2 Take-up re-randomization

Several benefit programs require a stochastic take-up decision: whether an eligible household actually participates. To construct the matrix correctly under take-up uncertainty, the pipeline runs three simulations per state: (a) a baseline simulation with original take-up values; (b) an all-take-up-true simulation that computes each entity's benefit value assuming full participation; and (c) a would-file-false simulation for tax unit variables.

The take-up rate for each program is resolved at the state level from the clone's assigned census block. Programs with take-up re-randomization include SNAP, ACA premium tax

credits, Medicaid, SSI, TANF, Head Start, and Early Head Start. The pipeline draws take-up booleans per (variable, household, clone) triple using a deterministic seeded random number generator. The seed is derived from the variable name, original household ID, and clone index, ensuring that draws are reproducible and independent across clones. Different clones of the same household may have different take-up draws, reflecting the fact that the same household placed in a different geographic area would face a different local take-up rate.

4.2.3 Clone loop and COO assembly

After all per-state simulations and take-up preparation complete, a clone loop iterates over each of the N clones. For each clone c:

1. Read the geographic slice: which state and district each record maps to in clone c.
2. Draw per-clone take-up booleans for each benefit program using the clone's geographic assignment and the deterministic seed.
3. Assemble per-record values by looking up the appropriate state simulation results for each record's assigned state. For take-up-dependent variables, entity-level values from the all-take-up-true simulation are multiplied by the per-clone take-up draw to produce the matrix entry for that clone.
4. Evaluate domain constraints—predicates that determine whether a record contributes to a particular target (e.g., "SNAP > 0" for SNAP dollar targets, "age ≥ 18" for adult population counts).
5. Emit nonzero entries as coordinate (COO) triples $(\text{row}, \text{col}, \text{value})$, where $\text{col} = c \times n_{\text{records}} + i$ for record i in clone c.

The COO entries from all clones are concatenated and converted to a compressed sparse row (CSR) matrix. Typical density is below 1%, as most records contribute to targets only in their assigned geographic area. This ordering ensures that the matrix accurately represents what the calibrated weights will reproduce when applied to the final dataset.

4.2.4 Target configuration

After the full unfiltered matrix is built, a configuration file (`target_config.yaml`) selects the subset of targets used for optimization. Each rule specifies a variable name, geographic

level, and optionally a domain variable. This separation allows the same expensive matrix to be reused with different target configurations without rebuilding.

Targets with zero row sums—meaning no record can contribute—are marked as unachievable and excluded from the loss function.

4.3 Stage 3: L_0-regularized weight optimization

4.3.1 Loss function

Given the sparse calibration matrix $\mathbf{M}$, a vector of target values $\mathbf{t} \in \mathbb{R}^m$, and a weight vector $\mathbf{w} \in \mathbb{R}^n$, the optimizer minimizes:

$$\mathcal{L}(\mathbf{w}, \boldsymbol{\alpha}) = \frac{1}{m} \sum_{j=1}^{m} \left(\frac{\hat{t}_j - t_j}{|t_j|} \right)^2 + \lambda_{L_0} \sum_{i=1}^{n} \bar{z}_i + \lambda_{L_2} \|\mathbf{w}\|_2^2 \tag{12}$$

where the weighted estimate for target j is:

$$\hat{t}_j = \sum_{i=1}^{n} M_{ji} \cdot w_i \cdot z_i \tag{13}$$

$z_i \in [0, 1]$ is the Hard Concrete gate for record i (Equations 1–4 in Section 2), $\bar{z}_i = \sigma(\log \alpha_i - \beta \log(-\gamma/\zeta))$ is the expected gate activation used as the L_0 penalty, λ_{L_0} controls the sparsity-accuracy trade-off, and λ_{L_2} provides mild weight regularization.

The calibration loss uses *relative* errors: a 1% miss on a $1 billion SNAP target receives the same gradient signal as a 1% miss on a $9 trillion employment income target. This scale-invariant formulation prevents large-magnitude targets from dominating the loss.

4.3.2 Hyperparameters

Table 2 lists the optimization hyperparameters with their values and roles. The stretch parameters $\gamma = -0.1$ and $\zeta = 1.1$ follow the original Hard Concrete paper, placing approximately 9% of the sigmoid's mass below 0 and above 1, which is what allows clipping to produce exact zeros and ones.

4.3.3 Initialization

Weights are initialized from age-bin population targets rather than uniformly. For each congressional district, the sum of age-bin target values gives the district's total population. The

Parameter	Value	Role
β (gate temperature)	0.35	Sharpness of 0/1 gate transition
γ (left stretch)	-0.1	Enables exact-zero gates after clipping
ζ (right stretch)	1.1	Enables exact-one gates after clipping
Initial keep probability	0.999	All records start nearly fully active
λ_{L_2}	10^{-12}	Mild weight regularization
Learning rate	0.15	Adam optimizer step size
Clone count N	430	Geographic replicas per CPS household

Table 2: Hard Concrete gate and optimization hyperparameters. The stretch parameters γ and ζ follow Louizos et al. (2018); other values are tuned for the calibration setting.

Source: `unified_calibration.py` in the `policyengine-us-data` repository.

initial weight for each record assigned to that district is set to the district population divided by the number of active records in the district. This gives the optimizer a demographically grounded starting point: records in high-population districts begin with higher weights. Small Gaussian jitter in log-weight space ($\sigma = 0.05$) and log-alpha space ($\sigma = 0.01$) breaks symmetry between duplicate CPS records.

Gate logits are initialized so that $P(z_i > 0) \approx 0.999$—nearly every record starts active. The optimizer begins from a dense, approximately calibrated state and prunes as the L_0 penalty accumulates over training epochs.

4.3.4 Presets

Two presets control the sparsity-accuracy trade-off via λ_{L_0}:

Preset	λ_{L_0}	Epochs	Retained records	Use case
National	10^{-4}	4,000	$\sim$50,000	Web-based interactive simulation
Local	10^{-8}	100	$\sim$3–4 million	Subnational policy analysis

Table 3: Sparsity presets. The only difference between presets is λ_{L_0} and epoch count; all other hyperparameters (Table 2) are shared.

Note: Retained record counts are approximate and vary with the base CPS size and clone count.

The only difference between presets is λ_{L_0}. All other hyperparameters are shared. A higher λ_{L_0} increases the gradient signal pushing gate logits below zero, pruning more records.

4.3.5 Optimization loop

The optimizer uses the Adam algorithm (Kingma and Ba, 2015) with learning rate 0.15. At each epoch:

1. Sample Hard Concrete gates z_i (stochastic during training).
2. Compute effective weights: $w_i^{\text{eff}} = \exp(\log w_i) \cdot z_i$.
3. Compute predictions: $\hat{t}_j = \sum_i M_{ji} \cdot w_i^{\text{eff}}$.
4. Evaluate the loss (Equation 12).
5. Backpropagate and update $\{\log w_i, \log \alpha_i\}$ via Adam.

At inference, stochastic gates are replaced by their deterministic counterparts. Records where $z_i^{\text{det}} = 0$ are excluded from the output dataset.

The current implementation runs for a fixed number of epochs (100 for regional fits, 4,000 for national fits) without early stopping. Convergence is assessed post hoc from diagnostic outputs.

4.4 Stage 4: weight expansion and dataset assembly

The optimizer returns a flat weight vector of length $N_{\text{clones}} \times n_{\text{records}}$. Stage 4 expands this into complete H5 dataset files suitable for microsimulation.

4.4.1 Weight reshaping and geographic filtering

The flat vector is reshaped to a matrix $\mathbf{W} \in \mathbb{R}^{N_{\text{clones}} \times n_{\text{records}}}$. Most entries are zero. For per-area datasets (e.g., a state dataset for California), clones whose congressional district falls outside the target area are zeroed out. For city datasets (e.g., New York City), clones whose county FIPS is not in the city's set of counties are zeroed out. The nonzero entries—typically 3–4 million for the local preset or approximately 50,000 for the national preset—identify the active (clone, household) pairs.

4.4.2 Entity membership preservation

Each CPS household contains persons who belong to sub-entities: tax units, SPM units, families, and marital units. The assembly step looks up the membership mapping for each active household and concatenates the corresponding person and sub-entity indices across

all active clones. Entity IDs are reassigned to globally unique values using a compound key (clone_id $\times$ offset $+$ old_id) with binary search remapping, achieving $O(n \log n)$ performance.

4.4.3 Geographic variable derivation

Each active clone carries a census block GEOID from the geography assignment. The assembly step derives state FIPS, county, tract, CBSA, SLDU, SLDL, place, VTD, PUMA, and ZCTA from the block GEOID. Because many clones share the same block (especially in dense urban areas), block GEOIDs are deduplicated before derivation and results are broadcast back via an inverse index.

4.4.4 SPM threshold recalculation

The Supplemental Poverty Measure threshold depends on local housing costs via a geographic adjustment factor. Because each clone's geography differs from the base dataset, SPM thresholds are recomputed using:

$$\text{threshold}_k = \text{base_threshold}(\text{tenure}_k) \times \text{equiv}(n_{\text{adults}}, n_{\text{children}}) \times \text{geoadj}(\text{CD}_k) \quad (14)$$

where the geographic adjustment factor for each congressional district is derived from median two-bedroom rent ratios from the Census Bureau.

4.4.5 Take-up consistency

The H5 assembly must produce identical take-up draws as the matrix builder for every (variable, household, clone) triple. Both stages call the same function (`compute_block_takeup_for_entities`) with the same seed derived from (variable name, original household ID, clone index). Any divergence would mean the matrix targeted a different subpopulation than what appears in the final dataset.

5 Results

This section presents calibration accuracy, sparsity analysis, convergence behavior, and comparisons with alternative methods. All empirical values are marked as placeholders pending completed pipeline runs.

5.1 Calibration accuracy by geographic level

Table 4 reports mean, median, and maximum absolute relative error across achievable targets, disaggregated by geographic level.

Geographic level	Targets	Mean ARE (%)	Median ARE (%)	Max ARE (%)
National	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
State	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
Congressional district	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**
All achievable	**[TBC]**	**[TBC]**	**[TBC]**	**[TBC]**

Table 4: Calibration accuracy by geographic level, measured as absolute relative error (ARE) across achievable targets under the local preset ($\lambda_{L_0} = 10^{-8}$).

Note: **[TBC: Values to be populated from** `unified_diagnostics.csv` **after a completed pipeline run.]**

The mean absolute relative error across all **[TBC: total count]** achievable targets is **[TBC: mean error]**%. At the national level, where the optimizer has the most flexibility (every record contributes), error is lowest at **[TBC: national mean]**%. State-level error averages **[TBC: state mean]**%, and congressional district error averages **[TBC: district mean]**%.

5.1.1 Error by target source

Error varies systematically across target sources. IRS SOI income targets, which have large magnitudes and strong signal in the calibration matrix, achieve mean error of **[TBC: IRS mean]**%. ACS age-band targets, which are count-based and distributed across 18 narrow bins per district, achieve mean error of **[TBC: ACS mean]**%. SNAP household count targets, which depend on the take-up re-randomization mechanism, achieve mean error of **[TBC: SNAP mean]**%.

5.1.2 Unachievable targets

Of the approximately 37,800 targets, **[TBC: count]** are marked unachievable (row sum zero in the calibration matrix). These correspond to congressional districts where no clones carry nonzero values for the target variable. Increasing the clone count from 430 reduces the number of unachievable targets, at the cost of a larger calibration matrix.

5.2 Sparsity analysis

The L_0 penalty produces exact zeros in the weight vector. Table 3 reports the number of retained records under each preset.

Under the national preset ($\lambda_{L_0} = 10^{-4}$), the optimizer retains approximately **[TBC: national count]** records from the initial **[TBC: total]** million, a reduction of **[TBC: national pct]**%. Under the local preset ($\lambda_{L_0} = 10^{-8}$), approximately **[TBC: local count]** million records are retained (**[TBC: local pct]**% reduction). The national preset provides a compact dataset suitable for in-browser simulation; the local preset preserves geographic resolution across all 436 congressional districts.

5.2.1 Weight distribution

Under the national preset, the weight distribution spans from **[TBC: min weight]** to **[TBC: max weight]**, with a median of **[TBC: median weight]**. The effective sample size (ESS), computed as $(\sum w_i)^2 / \sum w_i^2$, is **[TBC: ESS]** under the national preset, representing **[TBC: ESS pct]**% of the nominal sample size.

5.3 Convergence behavior

Figure 2 plots the mean absolute relative error across all achievable targets as a function of training epoch for both presets.

[Convergence curves to be generated from calibration_log.csv]

Figure 2: Mean absolute relative error over training epochs for the national preset (4,000 epochs, $\lambda_{L_0} = 10^{-4}$) and the local preset (100 epochs, $\lambda_{L_0} = 10^{-8}$).

Under the national preset, the mean error decreases from **[TBC: initial]**% at epoch 0 to **[TBC: final]**% at epoch 4,000. The error curve flattens by approximately epoch **[TBC: plateau epoch]**, suggesting that the remaining error reflects structural limitations of the calibration matrix rather than insufficient training. Under the local preset, the mean error decreases from **[TBC: local initial]**% to **[TBC: local final]**% over 100 epochs.

5.3.1 Residual error analysis

Targets with persistent high error after convergence fall into two categories. First, targets in sparsely populated congressional districts where few clones have nonzero matrix entries—the optimizer has insufficient degrees of freedom. Second, targets where the domain con-

straint is restrictive (e.g., EITC by number of qualifying children in a specific district), limiting the pool of contributing records.

5.4 Comparison with alternative methods

Table 5 compares the L_0 method with iterative proportional fitting (IPF) and generalized regression (GREG) estimators applied to the same calibration matrix and targets.

Method	Mean ARE (%)	Median ARE (%)	Runtime	Sparsity control
L_0 (this paper)	**[TBC]**	**[TBC]**	**[TBC]**	Yes (configurable λ_{L_0})
IPF	**[TBC]**	**[TBC]**	**[TBC]**	No
GREG	**[TBC]**	**[TBC]**	**[TBC]**	No

Table 5: Comparison of L_0 regularization with iterative proportional fitting (IPF) and generalized regression (GREG) estimators on the same calibration matrix and target set.
Note: **[TBC: Results to be populated after running IPF and GREG baselines. ARE = absolute relative error across all achievable targets.]**

[TBC: Comparison results to be populated after running IPF and GREG baselines on the same target set. Expected advantages of L0: joint multi-level calibration in a single pass, configurable sparsity, scalability to 37,800 targets. Expected disadvantages: longer computation time (GPU-based gradient descent vs closed-form or iterative solutions), sensitivity to hyperparameter selection.]

5.5 Application: subnational policy impact

To demonstrate the practical utility of the calibrated subnational datasets, we analyze the distributional impact of a policy reform at the state and congressional district level using POLICYENGINE's microsimulation engine.

[TBC: Policy application to be selected and simulated. Candidate: analysing EITC expansion impacts across congressional districts, showing how district-level datasets enable geographic disaggregation of winners and losers that national datasets cannot provide.]

6 Discussion

6.1 Configurable sparsity trade-off

The λ_{L_0} parameter provides a continuous sparsity-accuracy dial. At one extreme, $\lambda_{L_0} = 10^{-4}$ produces a compact dataset of approximately 50,000 records that loads in seconds for web-based simulation tools, where users expect near-instant feedback. At the other extreme, $\lambda_{L_0} = 10^{-8}$ retains 3–4 million records with fine geographic resolution across all 436 congressional districts. Intermediate values produce datasets of intermediate size.

This trade-off does not exist in classical calibration methods. IPF and GREG produce a single set of weights without sparsity control. To reduce dataset size, researchers must discard records post hoc or apply ad hoc thresholding—neither of which jointly optimizes accuracy and sparsity. The Hard Concrete gate provides a principled mechanism for this joint optimization, with λ_{L_0} serving as the researcher's preference parameter over the Pareto frontier.

This feature also broadens the method's practical role. The local preset yields area-specific unit-record files that resemble synthetic spatial microdata, while remaining anchored in observed survey households, model-based enhancements, and calibrated administrative totals. The paper's empirical evaluation remains focused on calibration baselines, but the resulting datasets are relevant to some of the same downstream use cases as synthetic-population workflows.

6.2 Computational cost

The pipeline runs on Modal, a cloud compute platform, using T4 GPUs for the optimization step. Stage 1 (clone creation and imputation) requires approximately 2–3 hours of CPU time. Stage 2 (matrix construction) requires approximately 2–3 hours across parallel workers, dominated by running POLICYENGINE simulations for each of the 51 state-level configurations. Stage 3 (optimization) requires approximately 30–60 minutes of GPU time for the national preset (4,000 epochs) and 5–15 minutes for the local preset (1,000 epochs). Stage 4 (H5 assembly) requires approximately 4–5 hours across parallel workers for all 488 (436 CDs, 50 states plus DC, NYC and a national) H5 builds.

Total wall-clock time for a complete pipeline run is approximately 8–12 hours. This exceeds the cost of a single IPF or GREG calibration, which takes minutes. However, the pipeline amortizes the expensive matrix construction step: once built, the calibration package can be reused with different λ_{L_0} presets or target configurations without rebuilding.

6.3 Limitations

6.3.1 Temporal gap between PUF and CPS

The most recent publicly available PUF references tax year 2015 (Bryant, 2023). The CPS ASEC is updated annually. Imputing 2015 tax return variables onto 2024 CPS records introduces a temporal mismatch. The QRF imputation conditions on demographic and income variables that are available in both datasets, partially mitigating the gap, but structural changes in the tax code since 2015—such as the Tax Cuts and Jobs Act of 2017—may not be fully captured.

6.3.2 No early stopping

The current implementation runs for a fixed number of epochs without an automatic convergence criterion. Convergence is assessed post hoc from diagnostic outputs. An adaptive stopping rule based on validation loss on held-out targets (excluded from training via `target_config.yaml`) could reduce computation time and prevent overfitting to the training target set.

6.3.3 Target selection sensitivity

The choice of which targets to include in the optimization affects calibration accuracy on excluded targets. The `target_config.yaml` mechanism provides flexibility but also introduces a modeling decision: including too many targets may overfit the weights to specific administrative totals, while excluding targets leaves them uncalibrated. The current configuration was developed iteratively through validation against held-out targets, but a more systematic approach to target selection—analogous to feature selection in supervised learning—could improve robustness.

6.3.4 Take-up uncertainty

Programme take-up draws are stochastic, introducing noise into the calibration matrix entries for take-up-dependent targets. Different random seeds produce different take-up assignments, which produce different calibrated weights. The pipeline uses deterministic seeds for reproducibility but does not currently quantify the sensitivity of calibration accuracy to the take-up random seed.

6.3.5 Geographic resolution

The current pipeline calibrates at three levels: congressional district, state, and national. Finer geographic levels—counties, metropolitan areas, school districts—are feasible in principle but would require additional target data and a larger calibration matrix. The census block assignment already provides the geographic foundation; the constraint is the availability of administrative targets at finer levels.

6.4 Generalizability

The method is not specific to the United States or to any particular tax-benefit system. Any setting where (a) a household survey must be calibrated to (b) administrative targets at (c) multiple geographic levels is a candidate for L_0-regularized calibration. The UK Family Resources Survey calibrated to HMRC tax data across local authorities, or the EU Statistics on Income and Living Conditions calibrated to Eurostat regional benchmarks, are plausible applications. The `l0-python` package implements the Hard Concrete optimizer independently of the POLICYENGINE framework.

7 Conclusion

This paper presents an L_0 regularization method for calibrating microsimulation datasets to subnational administrative targets. The method jointly optimizes household weight magnitudes and sparsity using Hard Concrete gates, producing calibrated datasets that simultaneously match approximately 37,800 targets across congressional districts, states, and the nation.

The key technical contribution is the adaptation of L_0 regularization—originally developed for neural network pruning—to the survey calibration setting. Each household-geography combination receives a continuous weight and a stochastic binary gate. The gate's learnable logit and the weight magnitude are jointly optimized via gradient descent, with the L_0 penalty controlling sparsity. At inference, gates collapse to deterministic zeros and ones, producing a sparse dataset in which the number of retained records is a smooth function of the penalty parameter λ_{L_0}.

The configurable sparsity is practically useful: the same pipeline produces compact 50,000-record datasets for web-based interactive tools and detailed 3–4 million-record datasets for subnational policy analysis, without separate calibration procedures.

The pipeline builds on the CPS-PUF imputation methodology of Woodruff and Ghenis

(2024), extending it from national to subnational coverage through clone-and-assign geography, hierarchical target reconciliation, and take-up re-randomization. The full pipeline—from raw CPS to calibrated H5 datasets—runs in approximately 8–12 hours on cloud infrastructure and is fully open source.

The method is implemented in two open-source packages: the `l0-python` PyTorch package (https://github.com/PolicyEngine/l0-python), which provides the Hard Concrete optimizer, and the `policyengine-us-data` pipeline (https://github.com/PolicyEngine/policyengine-us-data), which implements the full four-stage calibration. Calibrated datasets are publicly available on HuggingFace (`policyengine/policyengine-us-data`).

Future work includes extending calibration to county-level targets as additional administrative data becomes available, implementing adaptive λ_{L_0} scheduling during training (analogous to learning rate schedules), incorporating an early stopping criterion based on held-out target validation, and applying the method to non-US microsimulation systems.

Conflict of interest

The authors declare no conflict of interest.

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Data and code availability

The calibration pipeline, including the L_0 optimizer, is open source at https://github.com/PolicyEngine/policyengine-us-data. The `l0-python` PyTorch package is available at https://github.com/PolicyEngine/l0-python. Calibrated datasets are hosted on HuggingFace at `policyengine/policyengine-us-data`. The IRS Public Use File requires a restricted-access agreement; all other data sources are publicly available.

Use of generative AI

The authors used Claude (Anthropic) for code assistance and manuscript drafting. All technical content, methodology, and analysis were designed and validated by the authors.

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A Optimization hyperparameters

Table 6 lists all hyperparameters used in the L_0 optimization with their default values, code-level names, and roles.

Parameter	Code name	Value	Role
Gate temperature	`BETA`	0.35	Controls sharpness of 0/1 gate transition
Left stretch	`GAMMA`	-0.1	Enables exact-zero gates after clipping
Right stretch	`ZETA`	1.1	Enables exact-one gates after clipping
Initial keep probability	`INIT_KEEP_PROB`	0.999	All records start nearly fully active
Weight jitter SD	`LOG_WEIGHT_JITTER_SD`	0.05	Log-space noise on weights at init
Logit jitter SD	`LOG_ALPHA_JITTER_SD`	0.01	Log-space noise on gate logits at init
L_2 penalty	`LAMBDA_L2`	10^{-12}	Mild weight regularization
Learning rate	`LEARNING_RATE`	0.15	Adam optimizer step size
Default epochs	`DEFAULT_EPOCHS`	100	Training iterations (local preset)
Default clones	`DEFAULT_N_CLONES`	430	CPS copies before optimization
λ_{L_0} (local)	`PRESETS[``local'']`	10^{-8}	Sparsity penalty for local-area datasets
λ_{L_0} (national)	`PRESETS[``national'']`	10^{-4}	Sparsity penalty for national dataset
National epochs	(pipeline default)	4,000	Training iterations (national preset)

Table 6: Complete hyperparameter table for L_0 optimization. Source: `unified_calibration.py`.

B Calibration target variables

The calibration pipeline draws targets from seven administrative sources across three geographic levels. The database contains 37,758 active targets in total. The following tables list every target domain included in the `policy_data.db` database, grouped by geographic level.

B.1 Congressional district targets (33,572)

Target domain	Type	Count
Census ACS S0101		
Person count by age band (18 bands $\times$ 436 CDs)	count	7,848
IRS SOI		
Person count by AGI bracket (9 bins $\times$ 436 CDs)	count	3,924
EITC dollars by qualifying children (4 bins $\times$ 436 CDs)	$	1,744
Tax unit count by qualifying children (4 bins $\times$ 436 CDs)	count	1,744
Aggregate AGI (unconditional, $\times$ 436 CDs)	$	436
21 income/deduction dollar totals, each with domain > 0 ($\times$ 436 CDs)	$	9,156
Tax unit count for each of the 21 domains ($\times$ 436 CDs)	count	9,156
Census ACS S2201		
SNAP household count ($\times$ 436 CDs)	count	436
		33,572

Table 7: Congressional district calibration targets (436 CDs). Each row is replicated across all 436 districts. IRS SOI provides paired dollar and count targets for each income/deduction domain.

B.2 State targets (4,080)

Target domain	Type	Count
Census ACS S0101		
Person count by age band (18 bands $\times$ 51 states)	count	918
IRS SOI		
Person count by AGI bracket (9 bins $\times$ 51 states)	count	459
EITC dollars by qualifying children (4 bins $\times$ 51)	$	204
Tax unit count by qualifying children (4 bins $\times$ 51)	count	204
Aggregate AGI (unconditional, $\times$ 51 states)	$	51
20 income/deduction dollar totals (domain > 0, $\times$ 51)	$	1,020
Tax unit count for each of the 21 domains ($\times$ 51)	count	1,071
USDA FNS SNAP		
SNAP spending ($\times$ 51 states)	$	51
SNAP household count ($\times$ 51 states)	count	51
CMS Medicaid		
Medicaid enrollment ($\times$ 51 states)	count	51
Census STC		
State income tax collections ($\times$ 51 states)	$	51
		4,080

Table 8: State-level calibration targets (50 states + DC). IRS SOI variables mirror the district structure. USDA provides both SNAP spending and household counts; CMS provides Medicaid enrollment.

B.3 National targets (106)

Target domain	Type	Count
Demographics (Census ACS + curated)		
Person count by age band (18 bands)	count	18
Person count by SSN card type (4 categories)	count	4
Person count: Medicaid enrolled	count	1
Person count: ACA PTC recipients	count	1
IRS SOI — domain-constrained aggregates		
AGI (unconditional)	$	1
EITC by qualifying children (0–3+)	$	4
Tax unit count by qualifying children (0–3+)	count	4
21 income/deduction dollar totals (domain > 0)	$	21
Tax unit count for each of the 21 domains	count	21
CBO budget projections		
SNAP, Social Security, SSI, unemployment comp., income tax	$	5
SSA benefit decomposition		
Retirement, disability, survivors, dependents	$	4
JCT tax expenditure estimates		
SALT ded., charitable ded., mortgage interest ded., medical expense ded., QBI ded., EITC	$	6
Healthcare spending (MEPS/NHEA/CMS)		
Medicaid, health insurance premiums, Medicare Part B, other medical expenses, OTC health	$	5
Housing, transfers, and other		
Rent, real estate taxes, housing subsidy, work/childcare expenses, TANF, alimony (paid + received), child support (paid + received), tip income, net worth	$	10
Retirement contributions (IRS SOI)		
Traditional IRA, Roth IRA contributions	$	2
		106

Table 9: National-level calibration targets. CBO, JCT, SSA, CMS, and Census values are curated from the cited administrative sources and stored in the ETL pipeline. Dollar values are inflation-adjusted to the calibration year.

C Algorithm pseudocode

Algorithm 1 presents the complete L_0-regularized calibration procedure.

Algorithm 1 L_0-regularized calibration with Hard Concrete gates

Require: Calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$, targets $\mathbf{t} \in \mathbb{R}^m$, initial weights $\mathbf{w}_0 \in \mathbb{R}^n$
Require: Hyperparameters: λ_{L_0}, λ_{L_2}, β, γ, ζ, learning rate η, epochs E
Ensure: Calibrated sparse weight vector $\hat{\mathbf{w}} \in \mathbb{R}^n$

Initialize $\log w_i \leftarrow \log w_{0,i} + \mathcal{N}(0, 0.05^2)$ for all i
Initialize $\log \alpha_i \leftarrow \text{logit}(0.999) + \mathcal{N}(0, 0.01^2)$ for all i
Initialize Adam optimizer with parameters $\{\log w_i, \log \alpha_i\}$ and learning rate η
for epoch $= 1$ to E **do**
    Sample Hard Concrete gates (training):
    for $i = 1$ to n **do**
        $u_i \sim \text{Uniform}(\epsilon, 1 - \epsilon)$
        $\bar{s}_i \leftarrow \sigma\left(\frac{\log u_i - \log(1-u_i) + \log \alpha_i}{\beta}\right)$
        $s_i \leftarrow \bar{s}_i(\zeta - \gamma) + \gamma$
        $z_i \leftarrow \min(1, \max(0, s_i))$
    end for
    $w_i^{\text{eff}} \leftarrow \exp(\log w_i) \cdot z_i$ for all i
    $\hat{t}_j \leftarrow \sum_i M_{ji} \cdot w_i^{\text{eff}}$ for all j
    $\mathcal{L}_{\text{cal}} \leftarrow \frac{1}{m} \sum_{j=1}^{m} \left(\frac{\hat{t}_j - t_j}{|t_j|}\right)^2$
    $\mathcal{L}_{L_0} \leftarrow \sum_{i=1}^{n} \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right)$
    $\mathcal{L}_{L_2} \leftarrow \sum_{i=1}^{n} w_i^2$
    $\mathcal{L} \leftarrow \mathcal{L}_{\text{cal}} + \lambda_{L_0} \mathcal{L}_{L_0} + \lambda_{L_2} \mathcal{L}_{L_2}$
    Backpropagate $\nabla_{\log w, \log \alpha} \mathcal{L}$
    Adam step
end for
Deterministic inference:
for $i = 1$ to n **do**
    $z_i^{\text{det}} \leftarrow \min(1, \max(0,\ \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma))$
    $\hat{w}_i \leftarrow \exp(\log w_i) \cdot z_i^{\text{det}}$
end for
return $\hat{\mathbf{w}}$